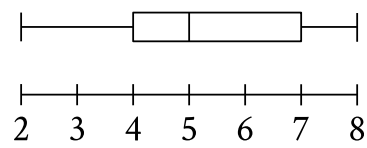


Question ID 57f45509

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 57f45509

PSAT1 M 2.1



The box plot summarizes **15** data values. What is the median of this data set?

- A. **2**
- B. **3**
- C. **5**
- D. **8**

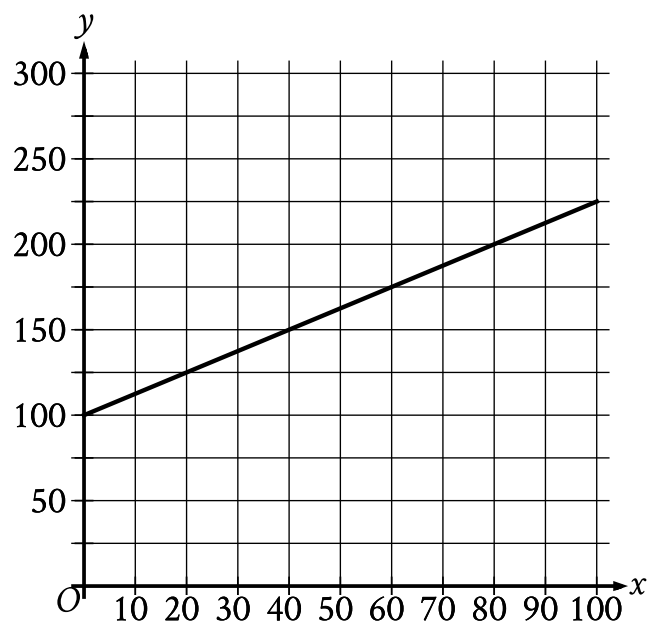
Question ID 720e51ac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 720e51ac

PSAT1 M 2.2

The cost y , in dollars, for a manufacturer to make x rings is represented by the line shown.



What is the cost, in dollars, for the manufacturer to make **60** rings?

- A. 100
- B. 125
- C. 175
- D. 225

Question ID b64e2c7f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: b64e2c7f

PSAT1 M 2.3

Monarch butterflies can fly only with a body temperature of at least **55.0 degrees Fahrenheit ($^{\circ}\text{F}$)**. If a monarch butterfly's body temperature is **51.3 $^{\circ}\text{F}$** , what is the minimum increase needed in its body temperature, in $^{\circ}\text{F}$, so that it can fly?

- A. **1.3**
- B. **3.7**
- C. **5.0**
- D. **6.3**

Question ID 837e9da7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 837e9da7

PSAT1 M 2.4

The function f is defined by $f(x) = \frac{1}{6x}$. What is the value of $f(x)$ when $x = 3$?

- A. $\frac{1}{3}$
- B. $\frac{1}{6}$
- C. $\frac{1}{9}$
- D. $\frac{1}{18}$

Question ID b4553284

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: b4553284

PSAT1 M 2.5

If $2x = 12$, what is the value of $9x$?

Question ID c3989ef8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: c3989ef8

PSAT1 M 2.6

Henry receives a **\$60.00** gift card to pay for movies online. He uses his gift card to buy **3** movies for **\$7.50** each. If he spends the rest of his gift card balance on renting movies for **\$1.50** each, how many movies can Henry rent?

- A. 10
- B. 25
- C. 35
- D. 40

Question ID ff2c1431

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: ff2c1431

PSAT1 M 2.7

$7m = 5(n + p)$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives n in terms of m and p ?

- A. $n = \frac{5p}{7m}$
- B. $n = \frac{7m}{5} - p$
- C. $n = 5(7m) + p$
- D. $n = 7m - 5 - p$

Question ID bf883fde

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: bf883fde

PSAT1 M 2.8

For the function f , the graph of $y = f(x)$ in the xy -plane has a slope of 3 and passes through the point $(0, -8)$. Which equation defines f ?

- A. $f(x) = 3x$
- B. $f(x) = 3x - 8$
- C. $f(x) = 3x + 5$
- D. $f(x) = 3x + 11$

Question ID 79201024

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 79201024

PSAT1 M 2.9

A band with **45** members has **11** members who play saxophone. If one band member is selected at random, what is the probability of selecting a band member who plays saxophone?

- A. $\frac{1}{45}$
- B. $\frac{11}{45}$
- C. $\frac{34}{45}$
- D. $\frac{45}{45}$

Question ID 1e003284

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 1e003284

PSAT1 M 2.10

$$\begin{aligned}x &= 49 \\ y &= \sqrt{x} + 9\end{aligned}$$

The graphs of the given equations intersect at the point (x,y) in the xy -plane. What is the value of y ?

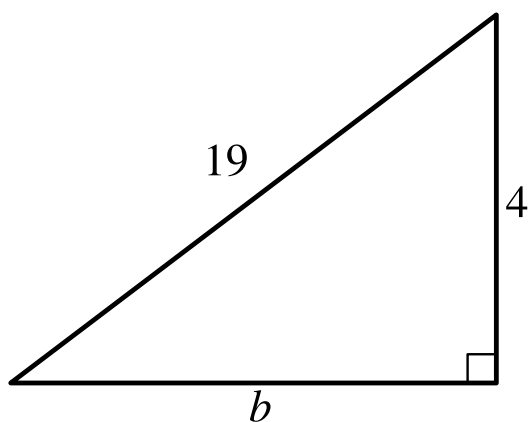
- A. 16
- B. 40
- C. 81
- D. 130

Question ID b0c5ece5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: b0c5ece5

PSAT1 M 2.11



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

- A. $4b = 19$
- B. $4 + b = 19$
- C. $4^2 + b^2 = 19^2$
- D. $4^2 - b^2 = 19^2$

Question ID c8e0f511

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: c8e0f511

PSAT1 M 2.12

For a camping trip a group bought x one-liter bottles of water and y three-liter bottles of water, for a total of **240** liters of water. Which equation represents this situation?

- A. $x + 3y = 240$
- B. $x + y = 240$
- C. $3x + 3y = 240$
- D. $3x + y = 240$

Question ID 2d16d62c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 2d16d62c

PSAT1 M 2.13

A special camera is used for underwater ocean research. When the camera is at a depth of **58** fathoms, what is the camera's depth in feet? (**1 fathom = 6 feet**)

Question ID e7343559

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: e7343559

PSAT1 M 2.14

$y = -4x + 40$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	0
1	-4
2	-8

B.

x	y
0	40
1	44
2	48

C.

x	y
0	40
1	36
2	32

D.

x	y
0	0
1	4
2	8

Question ID f9d40000

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: f9d40000

PSAT1 M 2.15

In $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . What is the measure of $\angle Z$?

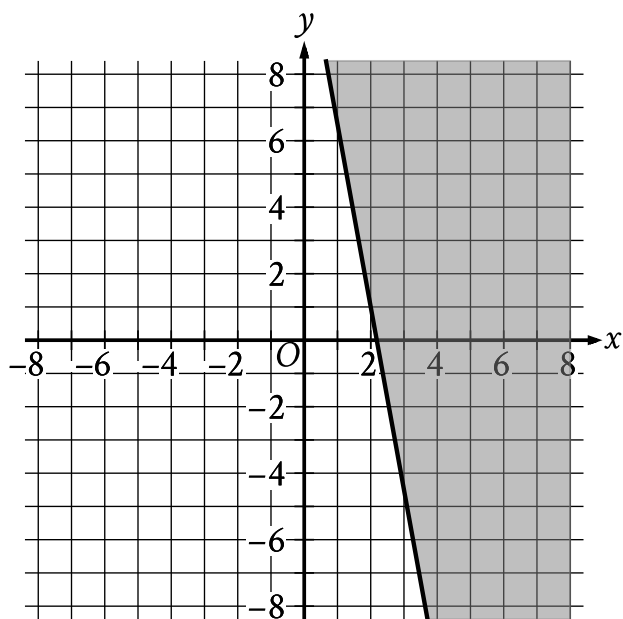
- A. 43°
- B. 89°
- C. 91°
- D. 179°

Question ID 59a49431

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	☒ ☐ ☐

ID: 59a49431

PSAT1 M 2.16



The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

- A. $(0, -4)$
- B. $(0, 4)$
- C. $(-4, 0)$
- D. $(4, 0)$

Question ID f88970cc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: f88970cc

PSAT1 M 2.17

$$\begin{aligned}x &= 5 \\ y &= x - 8\end{aligned}$$

Which of the following points (x, y) is the solution to the given system of equations in the xy -plane?

- A. $(0, 0)$
- B. $(5, -3)$
- C. $(5, -8)$
- D. $(5, 8)$

Question ID c7bed21d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c7bed21d

PSAT1 M 2.18

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P , Q , R , and S correspond to P' , Q' , R' , and S' , respectively. The measure of angle P is 30° , the measure of angle Q is 50° , and the measure of angle R is 70° . The length of each side of $P'Q'R'S'$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

- A. 10°
- B. 30°
- C. 40°
- D. 90°

Question ID 5ae186b4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 5ae186b4

PSAT1 M 2.19

$\frac{-54}{w} = 6$

What is the solution to the given equation?

Question ID d693f563

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: d693f563

PSAT1 M 2.20

Last year, Cedric had **35** plants in his garden. This year, the number of plants in Cedric’s garden is **60%** greater than the number of plants in his garden last year. How many plants does Cedric have in his garden this year?

Question ID 735a0a00

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 735a0a00

PSAT1 M 2.21

$$y = 0.25x^2 - 7.5x + 90.25$$

The equation gives the estimated stock price y , in dollars, for a certain company x days after a new product launched, where $0 \leq x \leq 20$. Which statement is the best interpretation of $(x, y) = (1, 83)$ in this context?

- A. The company's estimated stock price increased \$83 every day after the new product launched.
- B. The company's estimated stock price increased \$1 every 83 days after the new product launched.
- C. 1 day after the new product launched, the company's estimated stock price is \$83.
- D. 83 days after the new product launched, the company's estimated stock price is \$1.

Question ID 271ffad7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 271ffad7

PSAT1 M 2.22

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of **7** meters above the ground and reached a maximum height of **51.1** meters above the ground **3** seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of **7** meters?

- A. **3**
- B. **6**
- C. **7**
- D. **9**